

# Undergraduate Dissertation Research Proposal

## Solving American Option Free-Boundary Problems via Physics-Informed Neural Networks: Formulation, Stability, and Numerical Benchmarking

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**Proposed Area:** Scientific Machine Learning & Numerical Differential Equations  
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### Abstract

While European option valuation under the Black-Scholes model admits an exact analytical formula, American-style options allow early exercise at any time prior to maturity. This feature transforms the linear Black-Scholes Partial Differential Equation (PDE) into a non-linear **Parabolic Variational Inequality** (a free-boundary problem), where the optimal exercise boundary  $S^*(t)$  is an unknown moving curve. In this undergraduate research project, we develop a scientific machine learning framework to solve the American option pricing problem. We adopt a structured phased approach: (i) first developing and benchmarking a custom Physics-Informed Neural Network (PINN) against the classical Crank-Nicolson scheme with Projected Successive Over-Relaxation (PSOR) in 1D, and (ii) extending the mesh-free neural solver to multi-asset American basket options ( $d = 2, 5, 10$ ) where traditional grid-based finite difference methods suffer from the curse of dimensionality.

## 1 Problem Motivation & Mathematical Formulation

The classical Black-Scholes-Merton model (1973) assumes an asset price  $S$  following Geometric Brownian Motion. For **American options**, the holder retains the right to exercise at any time  $t \in [0, T]$ . Consequently, the option price  $V(S, t)$  must satisfy  $V(S, t) \geq h(S)$ , where  $h(S) = (K - S)^+$  is the intrinsic payoff for a put option with strike  $K$ . This yields a **Linear Complementarity Problem (LCP)**:

$$\min \left( -\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rS \frac{\partial V}{\partial S} + rV, \quad V(S, t) - h(S) \right) = 0, \quad (1)$$

subject to terminal and asymptotic boundary conditions:

$$V(S, T) = h(S), \quad \lim_{S \rightarrow \infty} V(S, t) = 0, \quad V(0, t) = K. \quad (2)$$

The domain naturally partitions into a *continuation region* (where the PDE holds with equality) and a *stopping region* ( $V = h(S)$ ), separated by the a priori unknown free boundary  $S^*(t)$ .

## 2 Brief Literature Survey & Research Gap

- **Classical Numerical Solvers:** Finite Difference Methods (FDM) discretize space-time grids. Brennan & Schwartz [3] and Cryer [4] developed the Projected Successive Over-Relaxation (PSOR) algorithm combined with implicit Crank-Nicolson schemes. While reliable in 1D, these mesh-based methods scale exponentially  $\mathcal{O}(N^d)$ , becoming computationally intractable for multi-asset baskets ( $d \geq 3$ ).
- **Physics-Informed Deep Learning:** Raissi et al. [1] formalized Physics-Informed Neural Networks (PINNs), embedding differential operators directly into loss functions via Automatic Differentiation (AD). Han et al. [2] showed the efficacy of deep learning in overcoming the curse of dimensionality for high-dimensional parabolic PDEs.
- **The Core Research Gap:** Standard PINNs struggle with variational inequalities due to non-differentiable kinks at  $S = K$  and the moving free boundary. Formulating smooth penalty functions and evaluating approximation errors against classical PSOR schemes offers a rigorous, publishable bridge between numerical analysis and deep learning.

## 3 Proposed Methodology

### 3.1 Phase 1: 1D Free-Boundary PINN vs. Crank-Nicolson PSOR (Core Baseline)

We construct a deep neural network  $V_\theta(S, t)$  parameterized by weights  $\theta$ . To handle the inequality constraint in a differentiable manner, we formulate a regularized penalty loss over collocation points  $\{(S_i, t_i)\}_{i=1}^{N_c}$ :

$$\mathcal{L}_{\text{total}}(\theta) = \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}}(\theta) + \lambda_{\text{early}} \mathcal{L}_{\text{early}}(\theta) + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}(\theta) + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}(\theta), \quad (3)$$

where the early-exercise penalty is given by  $\mathcal{L}_{\text{early}}(\theta) = \frac{1}{N_c} \sum_{i=1}^{N_c} [\max(0, h(S_i) - V_\theta(S_i, t_i))]^2$ .

**Numerical Benchmark:** We will implement a Crank-Nicolson PSOR solver as the ground-truth numerical baseline, evaluating:

- (i) Pointwise relative  $L_2$  and  $L_\infty$  error across the spatial-temporal domain.
- (ii) Accuracy of financial Greeks ( $\Delta = \frac{\partial V}{\partial S}$ ,  $\Gamma = \frac{\partial^2 V}{\partial S^2}$ ) computed via exact AD vs. numerical differentiation.
- (iii) Inference latency comparison between iterative PSOR and single forward-pass neural evaluation.

### 3.2 Phase 2: High-Dimensional Multi-Asset Extension ( $d = 2, 5, 10$ )

We extend the framework to  $d$ -asset basket options governed by the multidimensional coupled operator:

$$\mathcal{L}_d V \equiv -\frac{\partial V}{\partial t} - \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V}{\partial S_i \partial S_j} - r \sum_{i=1}^d S_i \frac{\partial V}{\partial S_i} + rV. \quad (4)$$

Because spatial discretization is eliminated, the PINN framework remains computationally tractable across dimensions  $d \in \{2, 5, 10\}$ , benchmarked against Longstaff-Schwartz Least Squares Monte Carlo (LSM).

## 4 Project Milestones & Timeline

| Term / Period     | Key Tasks & Milestones                                                 | Expected Deliverable                |
|-------------------|------------------------------------------------------------------------|-------------------------------------|
| <b>Months 1–2</b> | Mathematical derivation of LCP & implementation of 1D PSOR             | Verified FDM Baseline Code          |
| <b>Months 3–4</b> | Architecture design, custom penalty loss, & 1D PINN training           | Phase 1 Validation & Error Analysis |
| <b>Months 5–6</b> | Multi-asset formulation ( $d = 2, 5$ ), Greeks computation & stability | Phase 2 High-Dim Experiments        |
| <b>Months 7–8</b> | Comprehensive benchmarking, dissertation compilation, & drafting       | Final Thesis & Manuscript Draft     |

## References

- [1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
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- [5] F. Black and M. Scholes, “The pricing of options and corporate liabilities,” *Journal of Political Economy*, vol. 81, no. 3, pp. 637–654, 1973.